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FOUR STAR INDUSTRIES SINGAPORE – MATCHING SUPPLY WITH DEMAND

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In 2002, Neo Sia Meng took over as Executive Director of Four Star Industries Private Limited founded three decades earlier by his parents. After joining the family business as Director in 1996, Sia Meng saw several local and foreign mattress manufacturers enter the Singapore market. This intensified the competition in the local market which was already seasonal and volatile.

The increased competition resulted in a proliferation of product offerings from all the manufacturers. The local dealers who dominated the retail mattress market sought an exclusive range of in-house models from the local manufacturers. As Four Star depended heavily on the local dealers, it tried to accommodate their demands to increase its product range and service level. However, as Four Star substantially increased the variety of mattress offerings, its operations became chaotic, and matching demand with the correct inventory of mattresses became a significant challenge.

The order fulfillment problems created significant pressures on the manufacturing operations and Sia Meng was worried that these operational problems might provoke some of his long-serving employees to consider leaving the company. Sia Meng pondered on the short and long-term options for Four Star to move towards a responsive, but cost-effective operations model.

Professor S Viswanathan and D. G. Allampalli of Nanyang Business School prepared this case based on the interview of key personnel from Four Star Industries Private Limited. As the case is not intended to illustrate either effective or ineffective practices or policies, the information presented reflects the authors' interpretation of events and serves only to provide opportunities for classroom discussions. The authors would like to thank Mr Neo Sia Meng, Executive Director of Four Star Private Limited for the support in the development of this case study.

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THE V-SHOPPER ORDER

On 1st July 2002, Neo Sia Meng, Executive Director of Four Star Industries Private Limited discussed with his brother and Company Executive Neo Chiw Ern, the problems and chaos created by the unpredictability of demand which resulted in a significant decline in the order fill rates. Sia Meng had learnt from his Factory Manager that even after two weeks, the order of V-Shopper, a Four Star dealer, could not be completed due to stock-outs of certain slow-moving mattresses. The number of mattress models that Four Star manufactured had grown 18-fold in the previous six years. Though the warehouse space and inventory had increased substantially, the stock-outs of some mattress models resulted in the decline of order fill rates. Order fulfillment pressures were creating significant difficulty in production scheduling and putting increasing stress on manufacturing staff. Sia Meng was worried that these operational problems might provoke some of his long-serving employees to consider leaving the company.

BACKGROUND OF FOUR STAR INDUSTRIES

In 1966, Neo Gim Sin, a carpenter by profession, and his wife set up Four Star Industries Private Limited (Four Star) to produce spring mattresses. He chose mattress-making machines from Spühl AG, a leading European machinery manufacturer. Gim Sin recalled:

> *When we first started making mattresses in 1966, my wife and I were doing them by hand and we had to import bed springs from Hong Kong… Then I met a German mattress maker who invited me to visit his factory. In 1968, I visited his factory and several others in Europe. When I came back, I decided to buy Swiss machinery to make our mattresses.*[1]

Gim Sin bought the Swiss machines because he was impressed with their higher productivity and quality. As a result, Four Star's production rose 10-fold with the same number (five) of employees. Customer reactions to these mattresses were also very positive as they found them to be much better.[2]

In early eighties, Gim Sin observed that mattresses made using individually pocketed high tensile springs were gaining popularity as they provided better support for the back. He pioneered the introduction of this new product - *pocketed spring mattress* (PSM). Gim Sin first evaluated many Taiwanese machinery manufacturers, but eventually again chose the Swiss machinery again for their higher productivity. By 1985, he successfully introduced PSMs in Singapore.

Gim Sin's emphasis on quality and productivity yielded immediate dividends for Four Star. In 1988, it won the *Grand Prix for International Quality - Europe 89* from EDICON, a Madrid-based international trade magazine. Despite the quality recognition, Four Star did not enter the European markets due to high freight costs. Instead, it focused aggressively and exclusively on the local market. By the late eighties, Four Star had become a leading manufacturer of PSMs in Singapore and used a variety of spring counts in their construction. (See **Exhibit 1** – Four Star Models and Spring Counts in 1990s.) Its production had jumped from 500 mattresses[3] per annum in 1968 to 35,000 mattresses[4] in 1989.[5]

SINGAPORE MATTRESS MARKET

The Singapore mattress market in the 1990s comprised three major local manufacturers: Four Star, Dreammakers and Dunlopillo, a few foreign manufacturers, and traders. While Four Star and Dreammakers used their own technology, Dunlopillo made its mattresses through a franchise.[6] The leading foreign brands of mattresses in Singapore were Simmons and Slumberland. Simmons, an American brand manufactured its mattresses in Japan, while Slumberland had its manufacturing operations in Malaysia. Late entrants into the market included local brands such as Magic Koil, Wasaly and Prince Bed.

In early 2000, mattress manufacturers depended on hundreds of dealers, of which about 20 were large dealers (LDs) and the rest were small dealers (SDs).

1 Neo, L. S. (1989, December 25). A firm belief in technology pays off for Four Star. *The Business Times*, Singapore.
2 ibid.
3 ibid.
4 This figure included pocketed spring mattress as well as other types of mattresses such as bonnel spring mattress.
5 Local mattress maker to move into Asia-Pacific Market. (1989, December 16). *The Straits Times*, Singapore.
6 Chan, S, P. (1991, July 21). Dream of the dream mattress. *The Straits Times*, Singapore.

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LDs had huge showrooms in popular shopping malls. LDs such as Robinson, OG and METRO retailed a large variety of consumer products including mattresses while LDs like Novena and COURTS specialised in home and office furniture, and furnishing products. The SDs were spread throughout the HDB heartland[7] of Singapore, and sold mattresses and furniture to households in these public housing estates.

Together, LDs and SDs became the first point of contact for existing and potential customers. LDs sensitised the market, created the demand and facilitated new product launches. The dealers served customers who visited their showrooms and conducted seasonal promotions. They also offered credit card and loan facilities, and thus expanded their loyal customer base. The demand for mattresses was seasonal. However, the demand spiked significantly when the LDs provided steep discounts and conducted clearance sales. While the seasonal demand had a predictable pattern, for mattress makers, the order volumes were unpredictable during the highly discounted sales periods.

Before the late 1990s, LDs managed the variations in market demand by keeping inventory of mattresses in their own warehouses. Some LDs even maintained full-fledged delivery and logistics facilities to fulfill their customers' orders. However, towards the end of 1990s, LDs began to move away from these practices and by 2002, most LDs virtually kept no stock of mattresses, and expected mattress manufacturers to maintain the logistic facilities and provide direct delivery to their customers.

By 2002, the competition in the Singapore mattress market intensified. The LDs had a choice in terms of which mattress brands to push in their showrooms. At the same time, there was severe competition among the LDs themselves. The foreign mattress manufacturers also increased competitive pressures by launching several new products with technology-based materials and innovative designs. This led the LDs to broaden their product offerings. Many LDs therefore asked Four Star to develop in-house models[8] with higher value propositions. For example, Novena, Four Star's top dealer with 25 percent share of its sales, indicated to Sia Meng that it was unable to sell its spring-count based models. Based on Novena's request, Four Star developed an exclusive range of mattress models for the dealer, naming it *Premier* which comprised *Premier-Prestige, Premier-Luxurious, Premier-Firm, Premier-Optima* and *Premier-Deluxe*. The net effect of this was a proliferation in the number of mattress varieties manufactured by Four Star. (See **Exhibit 2 –** Number of New Models Introduced by Four Star.) Increased competition and a focus on higher-end mattress models contributed to a decrease in the total number of mattresses sold annually by Four Star. (See **Exhibit 3** – Production of Mattresses.)

PRODUCTION MANAGEMENT

When Sia Meng joined Four Star in 1996 and took over the responsibilities for product design and development, it had 13 mattress models. In response to increased competition, Sia Meng began developing new mattress models including in-house models for LDs. Four Star added 7 new mattress models in 1997, 60 models in 1998 and 40 in 1999. By 2002, Four Star manufactured 230 different models of spring mattresses. With the proliferation of new mattress models, and the resultant increase in finished goods inventory and warehouse capacity requirements, Four Star moved its factory to a building in an Industrial Estate in Kallang in the eastern part of Singapore. The manufacturing of pocketed spring mattresses at Kallang was organised into four departments: spring unit, quilt panel, assembly, and warehousing. (See **Exhibit 4** – Four Star – Manufacturing Process.)

Manufacturing Process

The manufacturing process of the pocketed spring mattress at Four Star began with the production of the innermost module, the spring unit.

Spring Unit: Four Star imported high tensile wires in three different gauges (1.8g, 2.1g and 2.4g). After testing the wires and coating them with anti-rust solvent, they were fed through a spring coiling machine to produce coil springs. Next, the springs were inserted into individual fabric pockets and sealed to produce pocketed springs. An optimum number of pocketed springs were then housed in a carbon steel frame to produce the spring unit assembly. The spring unit was the core chassis or

7 HDB (Housing and Development Board) heartland refers to the public housing estates where 84 percent of Singaporeans live, mostly in flats.

8 In-house models are mattress varieties that are sold exclusively at the retail outlets of a particular LD.

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frame of the mattress. There were different types of spring units used by Four Star, and these were due to the variations in the *spring count*[9] and the wire gauge used in the spring.

Quilt Panel: The second floor of the factory housed the quilt panel and assembly departments, as well as the warehousing facilities. Four Star imported a variety of fabrics in rolls of 400 metres from Belgian and German companies with plants in Indonesia and China, respectively. A foam sheet was inserted between two fabric layers and stitched into numerous patterns to produce a curvy outline on the top surface of the quilt. The quilt was then cut into panels of the appropriate size for the corresponding mattresses: *Single, Super Single, Queen* and *King*. For medium to high-end mattresses, Four Star used superior fabrics with additional finishing for water–resistant and ventilation properties.

Assembly: The mattress was assembled by wrapping the spring unit first with upholstery (i.e. foam and insulator layers), and then a quilt panel. Folding these layers manually, a tape end stitching machine (TESM) was used by Four Star's operators to complete the assembly and finishing of the mattresses.

Quality Control: After visual checks of the mattress, Four Star inspectors checked its feel and firmness, and those products that passed its quality standards were stored as finished goods inventory in its warehouse.

Production Scheduling

The capacity at each of the three major manufacturing stages (namely spring unit, quilt panel and assembly) was more than adequate to meet the current demand levels at Four Star. Only two of its machines had significant setup times; these were the coil spring machines and the quilt panel stitching machines. Each type of spring unit was produced in batches of 100.

The setup in the quilt panel stitching machine involved changing the cam and number of needles according to the sewing pattern for a particular quilt. The length of the fabric roll used for quilt panel production was 400 metres long, and this on the average yielded 50 quilt panels. Due to price and quality considerations, and the nature of the quilt stitching process, it was not practical for Four Star to use a fabric roll partly for one quilt. Hence the quilt panels were produced in batches of 50.

To produce the different types of mattresses, Four Star varied the number of pocketed springs, upholstery materials and quilt fabrics to adjust the feel, comfort, durability and the final price of a mattress. (See **Exhibit 5** – Raw Materials Composition of Fast-Moving Mattresses.) For example, in 2002, the mattress Model 'PU' was a Queen-sized mattress with a spring count of 788. The main components of this mattress model were the spring unit (with a spring count of 788), quilt fabric type 'QF1' and a foam layer of 25 mm thickness.

The springs were manufactured with wire of gauge 2.1 mm. The spring count in a mattress ranged from 384 to 1,308 to provide a wider choice to local dealers and customers. Similarly, 45 different quilt fabric rolls were used to create the 230 different mattress models.

Each fast-moving mattress model was assembled in batches of 9 to 12. To ensure that all three mattress assembly tables produced the same model at a time, the slow moving mattress models were assembled in batches of 3. Due to the large number of mattress varieties, fluctuations in the demand pattern, and the resultant production scheduling constraints, as well as service requirements, a safety stock corresponding to two weeks' requirement of all the different models of the finished mattresses was considered appropriate by Four Star.

In addition, the minimum order quantity (MOQ) for quilt fabric was between 2,000 to 3,000 metres. Four Star had to hold the cycle stock corresponding to the MOQ of all 45 different varieties of quilt fabrics it used. Similarly, the inventory of spring wire and foam material had to be held based on the corresponding MOQ for these raw materials. (See **Exhibit 6** – Minimum Order Quantities MOQ & Leadtimes for Raw Materials.)

As more LDs stopped keeping inventory of mattresses and required the manufacturer to perform the logistics and delivery functions, the service requirements imposed on Four Star increased significantly. Typically, the dealers expected Four Star to deliver the mattresses directly to its customers within a couple of days upon the placement of their orders.

9 The spring count (for a particular packing density of spring coils) is the number of spring coils used in a standard Queen-sized mattress with that coil packing density.

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The factory operations and sales departments of Four Star were headed by a Factory Manager and Sales Supervisor respectively, and both reported to Neo Chiw Ern. The Factory Manager was responsible for the entire production of mattresses and was supported by three supervisors and about 100 staff members who had been with the company for several decades. At the beginning of each week, Sia Meng, Chiw Ern, the Sales Supervisor and the Factory Manager respectively, met to decide on the production schedule.

Order Management

Both LDs and SDs placed their orders with Four Star after receiving a confirmed order from their retail customers. Their order volumes over the week-ends were higher compared to the weekdays, and remained very high and unpredictable during festive seasons and periods of discounted sales. The dealers used the fax or phone to convey their orders to Four Star. If adequate finished goods inventory of the ordered mattresses were available, then these orders were fulfilled by Four Star from existing stock. Otherwise, they were backordered, and manufactured on an expedited basis.

The order management and finished goods inventory planning were made difficult because the dealers did not plan or commit weekly or monthly purchases, and the responsibility of maintaining sufficient inventory of finished goods fell on Four Star. The dealers also expected Four Star to show a high degree of flexibility in accommodating the variations in mattress models, order sizes and delivery lead time despite the operational constraints. This, along with the minimum production batch size and high volatility of demand, made production scheduling and order fulfillment very complex.

Inventory and Demand Mismatch

In spite of the above factors, the order fulfillment process had worked well when Four Star sold only 13 mattress models. But with the proliferation of mattress varieties, the finished goods inventory levels had increased substantially. The warehouse storage space requirement increased from 5,000 square feet in 1998 to 45,000 square feet by 2002. At the same time, the order fill rates had declined to 60 percent by 2002 from 70 percent in the late 1990s. Sia Meng felt that the struggle to fulfill the V-shopper order was symptomatic of the operational problems faced by Four Star. Recounting the operational problems faced during 1999 to 2002, Sia Meng said:

> *The demand variability, dealers' expectations and insistence on in-house models with different quality and price necessitated overtime payments by 1999. In the next two years, the Factory Manager found it a challenge to coordinate shop floor activities and mistakes in production led to build up of months of inventory. Such mistakes caused wastages daily, and order fulfillment got delayed as stock outs became common.*[10]

Recalling the employees' frustration, he said:

> *Four Star began to lose good staff due to the chaotic operations and some of the very old employees expressed their intentions to quit. Some of them complained about the problems faced as neither the production nor the order management were going smoothly and caused stress to them. Some were undergoing treatment for management of blood pressure and even brought the medical report from their doctors to show that their health was affected due to the very stressful operational environment at work.*[11]

Sia Meng could accept that the decrease in sales revenue from S$9 million in 1989 to S$6 million in 2002 was due to the increased level of competition in the Singapore mattress market. What he could not come to terms with was the fact that Four Star had incurred operating losses during the previous three years. Four Star had to manage their bottom line better, and Sia Meng had to find a way to smoothly manage its operations so that factory employees were not overly stressed out.

Sia Meng was considering several options with Chiw Ern. One possibility was to reduce the number of mattress models manufactured to the levels in 1996 or 1998. But he was not sure whether this was acceptable to the dealers who might just move their business away to other local manufacturers. The second option was to insist that the dealers ordered

10 Interview with the authors on 4 August, 2008.
11 ibid.

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a minimum number of mattresses corresponding to the batch size for each model for which they placed an order. However, as the dealers were not carrying stock anymore, it was unlikely that they would accept this condition. Besides he was not sure whether this would alleviate his order fulfillment woes. Instead the whole distribution channel might end up carrying much higher inventory. A third possibility was to consider reducing the safety stock requirement for the finished mattresses, but Sia Meng was not sure how this could be done. Relocating the manufacturing to a cheaper overseas location was another possibility, but Sia Meng felt that this was not going to solve Four Star's problems, unless the fundamental issues related to production planning and inventory management were resolved. Though the problems appeared insurmountable, Sia Meng still felt that he could streamline his manufacturing, production scheduling, inventory management and order fulfillment processes to bring the company back into the black.

Sia Meng and Chiw Ern pored over the data available to them for an in-depth analysis. The demand variability for the 20 fast-moving mattresses was reasonably low, but for the slow-moving mattresses, the variability in relation to the average demand, was high. (See **Exhibit 7** – Demand Mean and Variance for some Fast-Moving Mattresses.) For the purpose of cost analysis, Sia Meng felt that he could assume that each of the remaining 210 models had an average annual demand of 30 mattresses and a standard deviation of 25. To simplify the analysis, he could also assume that bill of materials (BOM) for these models were similar to that for the 20 fast-moving mattresses (i.e. approximately every 10 of the remaining models had the same BOM as one of the 20 fast-moving models). The average cost value[12] of a mattress was S$150, with the spring unit, quilt panel and foam layer accounting respectively for S$85, S$35 and S$30 of the value.[13] Based on the warehouse labour costs, the annual rental paid to Jurong Town Corporation (JTC) for the factory premises, the allocated costs for overheads and the interest rate charged by its banker for working capital facilities, Sia Meng estimated that his annual inventory carrying costs for every Singapore dollar of inventory held by Four Star would be 20 cents.

Sia Meng and Chiw Ern wondered how they could streamline Four Star's operations and make sure that the legacy created by their parents would survive for another generation.

12 Excluding overhead and other indirect costs.
13 Interview with the authors on 4 August, 2008.

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EXHIBIT 1

FOUR STAR MODELS AND SPRING COUNTS IN 1990S



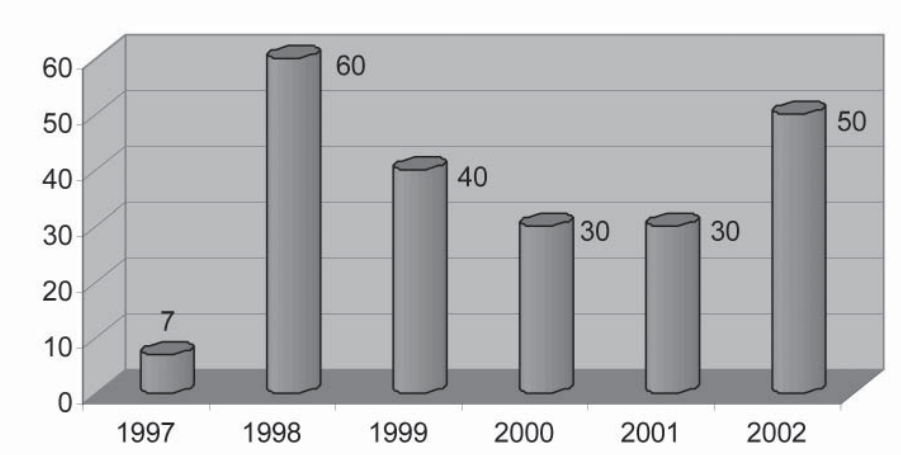
Source: Four Star spring mattresses. Retrieved September 16, 2009, from http://www.v-shopper.com.sg/vsales/default.htm?/vsales/mattress/fourstar/fourstar.htm

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EXHIBIT 2

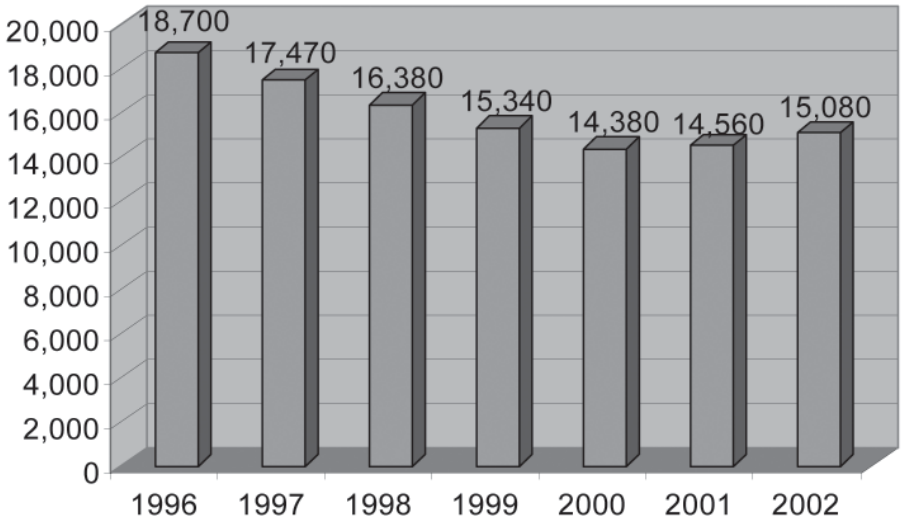
NUMBER OF NEW MODELS INTRODUCED BY FOUR STAR



Source: Four Star Industries Private Limited, Singapore.

EXHIBIT 3

PRODUCTION OF MATTRESSES



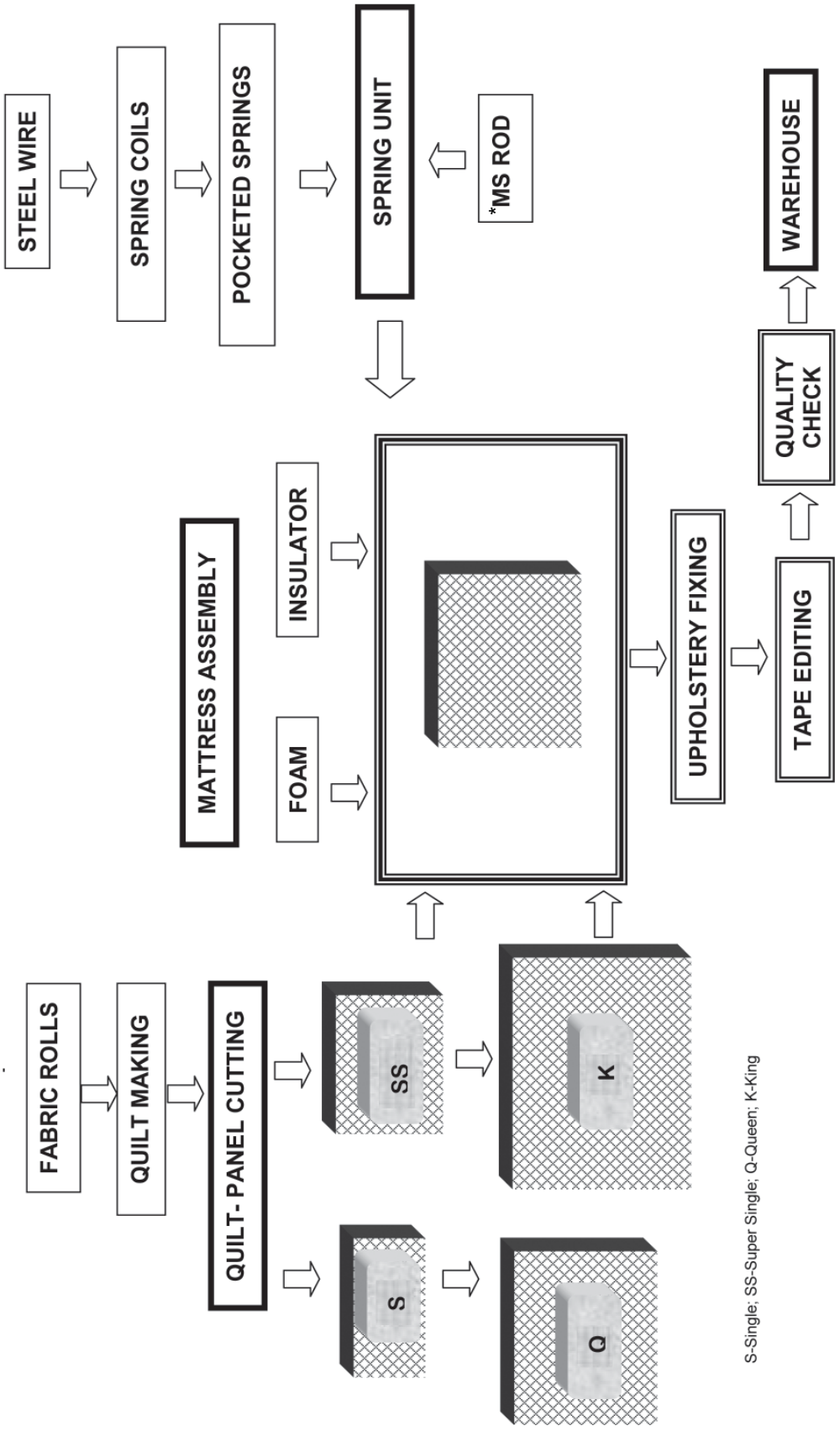
Source: Four Star Industries Private Limited, Singapore.

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EXHIBIT 4

FOUR STAR – MANUFACTURING PROCESS & DEPARTMENTS



S-Single; SS-Super Single; Q-Queen; K-King

* MS Rod refers to Mild Steel Rod.
Source: Four Star Industries Private Limited, Singapore.

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EXHIBIT 5

RAW MATERIAL COMPOSITION OF FAST-MOVING MODELS

VARIETY	DEMAND	SIZE	SPRING COIL COUNT	WIRE GAUGE	QUILT FABRIC	FOAM LAYER
PU	1,200	Q	788	2.1mm	QF1	25mm
CMD	920	Q	1,116	2.1mm	QF2	25mm
GOMK	800	Q	788	2.1mm	QF1	25mm
JC	650	Q	788	2.1mm	QF3	25mm
PL	630	Q	1,116	2.1mm	QF4	25mm
SM2	580	Q	788	2.1mm	QF4	18mm
SJL	540	Q	788	2.1mm	QF5	18mm
PU(S)	500	S	788	2.1mm	QF1	25mm
PF	480	Q	1,116	2.1mm	QF6	18mm
RC	440	Q	1,116	2.1mm	QF7	25mm
CMD(K)	380	K	1,116	2.1mm	QF2	25mm
OGN	330	Q	688	2.1mm	QF8	30mm
PU(K)	310	K	788	2.1mm	QF1	25mm
SBS	290	Q	688	2.1mm	QF8	25mm
GMK(K)	250	K	788	2.1mm	QF1	25mm
SC788	230	Q	788	2.1mm	QF1	25mm
GA	210	Q	788	2.1mm	QF9	30mm
USQ	200	Q	1,308	1.7mm	QF10	30mm
SC1116	190	Q	1,116	1.9mm	QF11	25mm
EX	170	Q	1,116	1.9mm	QF12	25mm
TOTAL	**9,300**					

Source: Four Star Industries Private Limited, Singapore.

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EXHIBIT 6

MINIMUM ORDER QUANTITIES (MOQ) & LEADTIMES (LT) FOR RAW MATERIALS

RAW MATERIALS		MOQ	PRICE/ UNIT	DELIVERY LT (WKS/DAYS)
SPRING WIRE	2.1mm	24,000kg	US$0.90	7 Weeks
	1.9mm	24,000kg	US$0.90	7 Weeks
QUILT FABRICS	QF1	3,000 metres	US$1.75	6 Weeks
	QF2	3,000 metres	US$2.35	6 Weeks
	QF3	2,000 metres	US$1.75	6 Weeks
	QF4	2,000 metres	US$2.35	6 Weeks
	QF5	2,000 metres	US$1.35	6 Weeks
	QF6	2,000 metres	US$2.00	6 Weeks
	QF7	2,000 metres	US$3.00	6 Weeks
	QF8	2,000 metres	US$2.70	6 Weeks
	QF9	2,000 metres	US$2.40	6 Weeks
	QF10	2,000 metres	US$3.50	6 Weeks
	QF11	2,000 metres	US$2.35	6 Weeks
	QF12	2,000 metres	US$4.20	6 Weeks
FOAM LAYER	7 mm	680 metres	S$2.18	2 Days
	12mm	680 metres	S$3.80	2 Days
	18mm	680 metres	S$4.77	2 Days

Source: Four Star Industries Private Limited, Singapore.

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EXHIBIT 7

DEMAND MEAN AND VARIANCE FOR SOME FAST-MOVING MATTRESSES

MATTRESS MODEL & SIZE	MEAN ANNUAL DEMAND μ	STANDARD DEVIATION OF ANNUAL DEMAND σ
PU-Single	500	38.8
PU-Queen	1,200	108.1
CMD-Queen	920	114.5
GOMK-Queen	800	102.9
JC-Queen	650	89.9
PL-Queen	630	61.7
SM2-Queen	580	53.1
SJL-Queen	540	43.6
PF-Queen	480	43.6
RC-Queen	440	34.2
OGN-Queen	330	30.8
SBS-Queen	290	24.3
Q788-Queen	230	18.5
GA-Queen	210	19.5
CLSQ-Queen	200	19.6
Q1116-Queen	190	17.3
EX-Queen	170	21.8
CMD-King	380	40.0
PU-King	310	38.3
GDMK-King	250	22.5

Source: Four Star Industries Private Limited, Singapore.

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